

Critical Point

① $F(x) = 2x = 0$

$x = 0 \rightarrow$ Critical Point

$F'(x) \Rightarrow DNE$
at $x = 0$

② $F(x) = \frac{1}{3}x^3 - 9x \quad [-7, 7]$

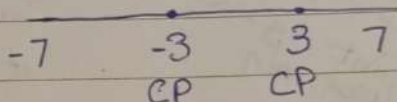
$\frac{1}{3}x^3 - 9x$

$F'(x) = x^2 - 9 = 0$

$x^2 = 9$

$x = \pm 3$

$F(-7) = \frac{1}{3}(-7)^3 - 9(-7)$
 $= -51.3$



$F(-3) = 18$

Abs max $\rightarrow 51.3$ at $x = 7$

$F(3) = -18$

Abs min $\rightarrow -51.3$ at $x = -7$

$F(7) = 51.3$

③ $F(x) = 1 - x^{2/3}$

$F'(x) = -\frac{2}{3}x^{-1/3} = 0$

$x^{-1/3} = -\frac{3}{2} = -0.29$

Critical Point (بعض intervals)
= Zero

Undefined

④ $F(x) = \frac{3\sqrt{x}}{x^2\sqrt{x+1}} \quad [-1, 1]$

$= 2x(\sqrt{x+1}) + x^2(\frac{1}{2\sqrt{x+1}})$

$(2x)(\sqrt{x+1}) + \frac{x^2}{2\sqrt{x+1}} = 0$ undefined

Crit. Point = -1, 0

$x+1 > 0$

$x > -1$

$(x+1)^{1/2}$
 $\frac{1}{2}(x+1)^{-1/2}$
 $\frac{1}{2\sqrt{x+1}}$

$F(0) = 1$

$F(-1) = (-1)^2(-1) = 0$

$F(1) = \sqrt{2}$

لوفى جذر بعض فى x
التحت بال zero

Local max, min

① $f(x) = 12x - x^2$

$f'(x) = -1 - 2x = 0$

$x = -\frac{1}{2}$

$f'(x) = 1 - 2x = 0 \rightarrow x = \frac{1}{2}$

$f'(1) = 1 - 2(1) = -1$

dec $0 - \frac{1}{2}$ inc $\frac{1}{2}$ dec

② $f(x) = \cos x$

$f'(x) = -\sin x = 0 \in [-\pi, \pi]$

$x = 0$

$x = -\pi$

$x = \pi$

Points: max, min

inc $-\pi$ to 0 dec 0 to π
In $[-\pi, 0]$ dec $[0, \pi]$

③

$x^2 - 3$; $[-3, 2]$

$f'(x) = 2x = 0 \rightarrow x = 0 \leftarrow \text{c.p.}$

Abs min, max

$f(0) = 3$

$f(-3) = 12$

$f(2) = 7$

$f'(-1) = -2$ dec $[-3, 0]$
 $f'(1) = 2$ inc $[0, 2]$

Local extrem min

Local extrem is local min or max

Inflection point & Concavity

$f(x) = 2x^3 - 3x^2 + 12$

$f'(x) = 6x^2 - 6x$

$f''(x) = 12x - 6 = 0$

$x = \frac{1}{2}$

$f(\frac{1}{2}) = 11.5$

$(\frac{1}{2}, 11.5)$

$f''(0) = -6 \rightarrow$ Concave down

$f''(1) = 6 \rightarrow$ Concave up

$\cap$ dec $\cup$